

Factories, Health, Safety and Welfare - The Foundation Track

Chapter F2.6 — From Zero: The Factories Act 1948, Hours and Leave, Occupational Disease, and the OSH Code

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NOTE · HOW TO USE THIS DOCUMENT

The Factories Act 1948 is a threshold-and-number statute, which makes it ideal examination material and ideal revision material. It is organised into chapters on health, safety and welfare, each triggered by a different employee count, plus a chapter on working hours and leave. The 2023 paper asked directly what is provided under the THIRD SCHEDULE - the list of notifiable occupational diseases - so that Schedule gets full treatment here. The OSH Code 2020 absorbed this Act along with twelve others and doubled the factory thresholds from 10 and 20 to 20 and 40. As a Mechanical Engineer you will find much of the safety content familiar; the value of this chapter for you is the NUMBERS, so it is built around them. Study time 6 hours across five sessions.

This is the FOUNDATION layer. Old-Act and Code provisions are given as pairs. One item is deliberately left blank - the Mines Act Special Officer notice period asked in 2023 - because it could not be verified against a primary source and will not be guessed at. See the note in Section 5.

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0. How to Study a Threshold Statute

This Act contains more numbers than any other in the module. Trying to remember them as a list will fail. Grouping them by what triggers them will not.

START FROM ZERO · THE ORGANISING INSIGHT - OBLIGATIONS SWITCH ON AT DIFFERENT HEADCOUNTS

The Factories Act does not impose one set of duties on all factories. It imposes a **ladder of duties**, each switching on at a different number of workers. So the way to learn it is:

not "what does the Act require?" - the list is too long

but "at **what number** does **this facility** become compulsory?"

Learn the **triggers**, and the facilities attach themselves. Section 3 of this chapter is built entirely as that trigger table, and it is the single most valuable page here.

STUDY SESSION · CHAPTER F2.6 - THE FIVE SESSIONS

- 1 **Session 1** - What a factory is, who the occupier is, and the approval machinery (Section 1)
- 2 **Session 2** - Health, safety and the hazardous-process provisions (Section 2)
- 3 **Session 3 - The welfare trigger table** - every facility and the headcount that compels it (Section 3)
- 4 **Session 4 - Working hours, overtime and leave** - the numbers chapter (Section 4)
- 5 **Session 5 - Occupational disease and the Third Schedule**, plus the OSH Code and the other twelve Acts (Section 5)

Then **45 graded questions**. Budget **six sessions**, about **three days**.

1. Session 1 - Factory, Occupier and Approval



1.1 What is a factory

MUST MASTER · THE DEFINITION, AND THE THRESHOLDS THE OSH CODE DOUBLED

A **factory** means any **premises** where:

- 1 **ten or more** workers are working, or were working on any day of the preceding twelve months, and a **manufacturing process** is being carried on **with the aid of power**, or
- 2 **twenty or more** workers, where the manufacturing process is carried on **without the aid of power**

Under the OSH Code, 2020 both figures doubled - twenty with power, forty without.

	Factories Act, 1948	OSH Code, 2020
With power	10 or more workers	20 or more workers
Without power	20 or more workers	40 or more workers

Expressly excluded from the definition: a **mine** subject to the Mines Act, a **mobile unit** belonging to the armed forces, a **railway running shed**, and a **hotel, restaurant or eating place**.

Note two things candidates miss. First, the phrase "**or were working on any day of the preceding twelve months**" - so a factory does not escape the Act by reducing headcount today. Second, the definition turns on **premises**, which includes the **precincts**, not merely the building.

JARGON DECODER · MANUFACTURING PROCESS - MUCH WIDER THAN MANUFACTURING

"**Manufacturing process**" is defined to include making, altering, repairing, ornamenting, finishing, packing, oiling, washing, cleaning, breaking up, demolishing or otherwise treating or adapting any article or substance **with a view to its use, sale, transport, delivery or disposal** - and also:

- 1 **Pumping** oil, water, sewage or any other substance
- 2 **Generating, transforming or transmitting power**
- 3 **Composing types for printing**, printing, bookbinding
- 4 **Constructing, reconstructing, repairing, refitting, finishing or breaking up ships or vessels**
- 5 **Preserving or storing any article in cold storage**

So **packing, cold storage** and **power generation** are all manufacturing processes. A question offering "cold storage" as something outside the Act is testing this. As an engineer you will find the breadth intuitive - the Act follows **hazard**, not commercial category.

1.2 Occupier

The **occupier** of a factory means the person who has **ultimate control** over its affairs. The Act then adds deeming provisions:

- 1 In the case of a **firm or other association of individuals**, any one of the **individual partners or members** is deemed to be the occupier
- 2 In the case of a **company**, any one of the **directors**
- 3 In the case of a factory owned or controlled by the **Central or State Government** or a local authority, the person or persons **appointed to manage** its affairs

IN PLAIN ENGLISH · WHY THE LAW NAMES A HUMAN BEING

A company cannot be imprisoned. If the Act simply made "the factory" liable, safety obligations would reduce to fines absorbed as a cost of business.

By deeming a **director** to be the occupier, the Act places the duty on a **named individual** who can be prosecuted personally. That is a deliberate design choice, and the same logic runs through the EPF & MP Act, where liability attaches to the employer including the **manager or occupier**, and where officers of a company can be proceeded against.

The lesson for you as an enforcement officer: **identify the individual on whom the statute places the duty**. Proceeding against an abstraction achieves nothing.

1.3 Approval, licensing and notice

- 1 The **State Government** may make rules requiring **previous permission** for the site, construction or extension of a factory, and the plans must be approved
- 2 Where an application for permission is **not decided within three months**, permission is **deemed granted**
- 3 The occupier must give the **Inspector** written **notice at least fifteen days before** he begins to occupy or use any premises as a factory
- 4 For a **seasonal** factory, notice is **at least thirty days** before work begins
- 5 A **Site Appraisal Committee** advises the State Government on applications for the initial location of a factory involving a **hazardous process**

CHECK YOURSELF · SESSION 1

- 1 Give the factory thresholds under the 1948 Act and under the OSH Code
- 2 Name four categories expressly excluded from the definition of factory
- 3 Why does the phrase "or were working on any day of the preceding twelve months" matter?
- 4 Name five activities that are manufacturing processes though not obviously manufacturing
- 5 Who is the occupier of a company-owned factory, and why does the Act name an individual?
- 6 Give the notice period before occupying a factory, and for a seasonal factory
- 7 What happens if permission for a site is not decided within three months?

2. Session 2 - Health, Safety and Hazardous Processes ★★★★★☆

2.1 Health provisions

Learn these as a checklist of eight, several with numbers attached:

- 1 **Cleanliness** - accumulations of dirt and refuse removed daily; floors cleaned weekly by washing or other effective method
- 2 **Disposal of wastes and effluents** - effective arrangements for treatment and disposal
- 3 **Ventilation and temperature** - adequate ventilation and reasonable temperature; walls and roofs of prescribed material where necessary
- 4 **Dust and fume** - effective measures to prevent inhalation and accumulation; no internal combustion engine exhaust in an enclosed space without ventilation
- 5 **Artificial humidification** - only with water of prescribed purity standards
- 6 **Overcrowding** - not less than **14.2 cubic metres** of space per worker. Note the qualification: in computing this, no space more than **4.2 metres** above floor level is taken into account
- 7 **Lighting** - sufficient and suitable, natural or artificial; glare and shadow to be prevented
- 8 **Drinking water** - effective arrangements at convenient points, legibly marked "**drinking water**", and at least **6 metres** away from any washing place, urinal or latrine. Where **more than 250 workers** are employed, provision for **cooled** drinking water during hot weather
- 9 **Latrines and urinals** - separately for male and female workers, adequately lighted and ventilated, kept clean. Where **more than 250 workers** are employed, they must be of **prescribed sanitary types** with glazed or polished surfaces
- 10 **Spittoons** - sufficient number, maintained clean and hygienic

MEMORY TECHNIQUE · THE HEALTH NUMBERS

14.2 cubic metres per worker, ignoring space above **4.2 metres** · drinking water **6 metres** from washing places and latrines · **cooled** drinking water and **sanitary-type** latrines above **250 workers**. Only four numbers in the whole health chapter, and **250** does double duty. That is manageable.

2.2 Safety provisions

- 1 **Fencing of machinery** - all dangerous parts securely fenced
- 2 **Work on or near machinery in motion** - only by a specially trained adult male worker wearing tight-fitting clothing, whose name is recorded in the prescribed register
- 3 **Employment of young persons on dangerous machines** - prohibited unless fully instructed as to the dangers, sufficiently trained, and under adequate supervision
- 4 **Striking gear and devices for cutting off power**
- 5 **Self-acting machines** - no traversing part to run within **45 centimetres** of any fixed structure that is not part of the machine
- 6 **Casing of new machinery, hoists and lifts, lifting machines, chains, ropes and lifting tackles, revolving machinery, pressure plant, floors, stairs and means of access, pits, sumps and openings in floors**
- 7 **Excessive weights** - no worker to lift, carry or move any load so heavy as to be likely to cause him injury
- 8 **Protection of eyes** - screens or goggles where there is risk of injury from particles or from exposure to excessive light

- 9 **Precautions against dangerous fumes and gases**, and against **explosive or inflammable dust and gas**
- 10 **Precautions in case of fire**
- 11 **Safety Officers** - required where **one thousand or more** workers are ordinarily employed, or where the manufacturing process involves risk of bodily injury, poisoning or disease

KNOW · HAZARDOUS PROCESSES - THE CHAPTER ADDED AFTER BHOPAL

The provisions on **hazardous processes** were inserted by the **1987 amendment**, in the aftermath of the **Bhopal gas disaster of 1984**. They are examined because they are conceptually distinct from the general safety provisions.

- 1 **Site Appraisal Committee** - advises on the initial location of a factory involving a hazardous process
- 2 **Compulsory disclosure of information** by the occupier to workers, the Chief Inspector, the local authority and the general public - the **right to know**
- 3 **Specific responsibility of the occupier** in relation to hazardous processes
- 4 **Medical examination** of workers, and appointment of persons with prescribed qualifications to handle hazardous substances
- 5 **Permissible limits of exposure** to chemical and toxic substances, specified in the **Second Schedule**
- 6 **Workers' participation in safety management** - a **Safety Committee** is required in every factory carrying on a hazardous process where **fifty or more** workers are ordinarily employed
- 7 **Right of workers to warn about imminent danger**

The First Schedule lists the **industries involving hazardous processes**. Keep the three Schedules distinct: **First = hazardous industries**, **Second = permissible exposure limits**, **Third = notifiable diseases**.

CHECK YOURSELF · SESSION 2

- 1 Give the overcrowding standard and the height qualification
- 2 How far must drinking water be from a latrine? What changes above 250 workers?
- 3 Who may work on or near machinery in motion, and what must be recorded?
- 4 At what headcount is a Safety Officer required?
- 5 Which event prompted the hazardous-process provisions, and in what year were they inserted?
- 6 At what headcount is a Safety Committee required in a hazardous-process factory?
- 7 What does each of the First, Second and Third Schedules contain?

3. Session 3 - The Welfare Trigger Table ★★★★★

This is the page to memorise. Every question on welfare provisions is a question about one of these numbers.

MUST MASTER · THE TRIGGER TABLE - WHAT BECOMES COMPULSORY AT WHAT HEADCOUNT

Trigger	Facility	Detail
Every factory	Washing facilities	Adequate and suitable, separately for male and female workers, screened
Every factory	Facilities for storing and drying clothing	And for sitting, for workers obliged to work standing
Every factory	First-aid appliances	One first-aid box per 150 workers , in the charge of a person trained in first aid
150 workers	Shelters, rest rooms and lunch rooms	Where more than 150 workers are ordinarily employed
250 workers	Canteen	Where more than 250 workers are ordinarily employed. Also cooled drinking water and sanitary-type latrines at this figure
500 workers	Ambulance room	An ambulance room with prescribed equipment and staff, where 500 or more workers are ordinarily employed
500 workers	Welfare Officer	Where 500 or more workers are ordinarily employed
1,000 workers	Safety Officer	Where 1,000 or more workers are ordinarily employed
30 women	Creche	Where more than thirty women workers are ordinarily employed - a room for children under six
50 workers	Safety Committee	In a factory carrying on a hazardous process with 50 or more workers

The sequence to chant: 150 rest rooms · 250 canteen · 500 ambulance room and Welfare Officer · 1,000 Safety Officer · 30 women creche · 150 workers per first-aid box.

MUST MASTER · THE CRECHE - AND HOW THE CODES CHANGED IT

The creche is the provision most often revised, so learn the pair.

- 1 **Factories Act, 1948** - creche required where more than **thirty women workers** are ordinarily employed
- 2 **Under the Codes and the 2026 Central Rules** - a creche is required in an establishment with **fifty employees**, and the criterion is **employees**, not women workers. The creche must be **within 500 metres** or as prescribed, and the Rules provide for a maintenance allowance of **Rs 500 per child** for up to **two children**

Note the two conceptual changes, both examinable:

- 1 From **thirty women** to **fifty employees** - the obligation no longer depends on how many **women** are employed. Childcare is treated as a **parental** need, not a women's need. That is a genuine shift in principle and worth a sentence at interview
- 2 The **Maternity Benefit Act**, as amended in **2017**, separately requires a creche in establishments with **fifty or more employees**, with **four visits a day** permitted to the mother. So the fifty-employee creche appears in **two** statutes - keep them both in mind

BEGINNER MISTAKE · THREE WELFARE TRAPS

- 1 **250 is the canteen, not the ambulance room.** The ambulance room is **500**. Candidates swap them constantly
- 2 **Welfare Officer at 500, Safety Officer at 1,000.** Welfare comes first because it is cheaper; safety expertise is required only in the largest plants. If you can remember which is more expensive to provide, you can remember which number is higher
- 3 **The first-aid figure is one box per 150 workers**, which is a **ratio**, not a threshold. Every factory needs first-aid provision regardless of size; the 150 governs **how many boxes**

CHECK YOURSELF · SESSION 3

- 1 Recite the trigger table from memory: 150, 250, 500, 1,000, 30 women, 50 hazardous
- 2 Which facility is required at 250 and which at 500?
- 3 Which officer is required at 500 and which at 1,000?
- 4 Give the creche requirement under the Factories Act and under the Codes, and state the two conceptual changes
- 5 Which other statute also requires a creche at fifty employees, and how many visits does it permit?
- 6 Is the 150 figure for first aid a threshold or a ratio?

4. Session 4 - Working Hours, Overtime and Leave



4.1 Adults - the hours provisions

Provision	Figure	Detail
Weekly hours	48	No adult worker to work more than forty-eight hours in a week
Weekly holiday	1	One day in a week, ordinarily Sunday. Substitution permitted within three days before or after , with notice, but no worker may work more than ten days consecutively without a holiday
Daily hours	9	No adult worker to work more than nine hours in a day
Rest interval	5	No period of work to exceed five hours without an interval for rest of at least half an hour
Spread-over	10.5	The periods of work inclusive of rest intervals must not spread over more than ten and a half hours in a day - extendable to twelve with the written permission of the Chief Inspector
Night shift	24 hours	Where a shift extends beyond midnight, the weekly holiday means twenty-four consecutive hours beginning when the shift ends
Overtime rate	Twice	Where a worker works more than nine hours a day or forty-eight hours a week, he is entitled to overtime at twice the ordinary rate of wages
Overtime cap	Quarterly	Total hours of overtime were capped per quarter; the 2026 Central Rules set the cap at 144 hours per quarter
Double employment	Prohibited	No adult worker to be required or allowed to work in any factory on any day on which he has already worked in another factory

Hours of work for adults - eleven numbers, and every one has been examined somewhere

MEMORY TECHNIQUE · THE HOURS NUMBERS AS A SINGLE STRING

48 week · 9 day · 5 hours then a half-hour break · 10.5 spread-over (12 with permission) · 1 weekly holiday · never more than 10 days consecutively · overtime at twice · 144 hours per quarter.

Say it as a sentence rather than a table and it will stay.

4.2 Young persons and women

- 1 A **child** below **fourteen** years may **not** be employed in any factory
- 2 An **adolescent** between **fourteen and eighteen** may be employed only if he has a **certificate of fitness** granted by a certifying surgeon and carries a **token** referring to it
- 3 A young person who has been granted a certificate of fitness and is **at least fifteen** may be treated as an **adult** for the purposes of hours of work, provided the certificate so states
- 4 Working hours for **children** are limited to **four and a half hours** a day, and they may **not work at night**
- 5 **Women** may not be required to work except between **6 a.m. and 7 p.m.** under the 1948 Act. **Under the Codes, women are permitted to work at night with their written consent**, subject to conditions on **transport, safety, adequate numbers** and compliance with the **POSH** requirements

WHY THIS EXISTS · NIGHT WORK FOR WOMEN - A RESTRICTION TURNED INTO A CHOICE

The 1948 position was **protective prohibition**: women simply could not be employed at night. The stated aim was safety; the effect was **exclusion** from shift work, promotion tracks and better-paid roles in continuous-process industries.

The Codes replace prohibition with **regulated consent**: a woman may work at night **if she consents in writing**, and the employer must supply **transport, safety arrangements, adequate numbers of women on the shift** and **POSH compliance**.

The principle: the answer to a risk faced by women is to **remove the risk**, not to remove the women. That is a good interview formulation, and it echoes the **Article 15(3)** point from Chapter F1.3 - special provision for women is permitted, but the courts have increasingly asked whether a given "protection" actually protects or merely excludes.

4.3 Annual leave with wages

MUST MASTER · THE LEAVE FORMULA - AND THE THREE NUMBERS

A worker who has worked for **240 days or more** in a calendar year is entitled to leave with wages in the following calendar year at the rate of:

- 1 **One day for every twenty days** of work performed, for an **adult**
- 2 **One day for every fifteen days** of work performed, for a **child**

Carry forward - unavailed leave may be carried forward to the succeeding year, up to a maximum of **thirty days** for an adult and **forty days** for a child.

Note the pattern: the **child** gets the **more generous** ratio (one in fifteen rather than one in twenty) and the **higher** carry-forward (forty rather than thirty). The Act is deliberately more protective of the younger worker.

The 240-day figure is the same as the continuous-service figure you learned in Chapter F2.3 for retrenchment compensation. **One number, two statutes** - a genuine economy in your revision.

Under the OSH Code the leave entitlement was made more generous - the qualifying requirement is **180 days** of work in a calendar year, with the accrual rate of **one day for every twenty days** retained.

CHECK YOURSELF · SESSION 4

- 1 Recite the hours numbers as a sentence
- 2 What is the spread-over limit, and to what may it be extended and by whom?
- 3 What is the maximum number of consecutive days a worker may work without a holiday?
- 4 Give the age limits for children and adolescents, and what an adolescent needs to be employed
- 5 State the 1948 position and the Code position on night work for women, and the principle behind the change
- 6 Give the leave formula for an adult and for a child, with the carry-forward limits
- 7 What is the qualifying number of days under the Factories Act and under the OSH Code?

5. Session 5 - Occupational Disease and the OSH Code ★★★★★

The 2023 paper asked what is provided under the Third Schedule. This session answers that and places the Act inside the new Code.

5.1 The Third Schedule - notifiable diseases

MUST MASTER · THE THIRD SCHEDULE CONTAINS THE LIST OF NOTIFIABLE OCCUPATIONAL DISEASES - ASKED IN 2023

The **Third Schedule** to the Factories Act, 1948 lists the **diseases** which, when contracted by a worker in a factory, must be **notified**.

The mechanism, which is what makes the Schedule meaningful:

- 1 Where any worker in a factory contracts a disease specified in the Third Schedule, the **manager** must send **notice** to the Chief Inspector in the prescribed form
- 2 Any **medical practitioner** who attends a person whom he believes to be suffering from such a disease must **report** to the Chief Inspector, and failure to do so attracts a penalty
- 3 The State Government may require **medical examination** and may appoint **certifying surgeons**

Diseases in the Third Schedule include:

- **Lead, tetraethyl lead, phosphorus, mercury, manganese, arsenic** and their compounds - poisoning and its sequelae
- **Carbon bisulphide, benzene** and its homologues, **chrome ulceration, chlorine, bromine, iodine, fluorine** and their compounds
- **Nitrous fumes** poisoning
- **Anthrax**
- **Silicosis**
- **Asbestosis**
- **Pathological manifestations due to radium, radioactive substances or X-rays**
- **Primary epitheliomatous cancer of the skin**
- **Toxic anaemia, toxic jaundice** due to poisonous substances
- **Occupational cataract** due to infra-red radiation
- **Compressed air illness**
- **Byssinosis**
- **Noise-induced hearing loss**
- **Occupational asthma, contact dermatitis, pesticide poisoning**

The three Schedules, one more time, because the exam swaps them:

- 1 **First Schedule - industries** involving **hazardous processes**
- 2 **Second Schedule - permissible limits of exposure** to chemical and toxic substances
- 3 **Third Schedule - notifiable diseases**

A note on the three most examined diseases and their causes, since your engineering background makes these easy: **silicosis** from free silica dust in mining, stone-crushing and foundry work; **asbestosis** from asbestos fibre; **byssinosis** from cotton dust in textile mills. Those three are the classic Indian occupational diseases and appear far more often than the rest.

5.2 Notice of accidents and dangerous occurrences

- 1 Where an accident causing **death** or **bodily injury** preventing the person from working for **forty-eight hours or more** occurs, the manager must send notice to the prescribed authorities
- 2 Where a **dangerous occurrence** of a prescribed kind occurs, notice must be given **whether or not** it causes bodily injury
- 3 The State Government may require the maintenance of an **accident register**

5.3 The OSH Code, 2020 - thirteen Acts in one

MUST MASTER · THE THIRTEEN ACTS, AND THE FOUR YOU MUST BE ABLE TO NAME

The **Occupational Safety, Health and Working Conditions Code, 2020** subsumes **thirteen** Acts - the largest of the four Codes by number of statutes replaced. Recall the **4-3-9-13** sequence.

The ones you must be able to name:

- 1 The **Factories Act, 1948**
- 2 The **Mines Act, 1952**
- 3 The **Contract Labour (Regulation and Abolition) Act, 1970**
- 4 The **Inter-State Migrant Workmen Act, 1979**
- 5 The **Building and Other Construction Workers Act, 1996**
- 6 The **Plantations Labour Act, 1951**
- 7 The **Dock Workers Act, 1986**
- 8 Plus the Acts on **working journalists, motor transport workers, beedi and cigar workers, sales promotion employees, cine workers** and the **Employment Exchanges** notification Act

Key changes made by the OSH Code:

- 1 **Factory thresholds doubled - 20 with power, 40 without**
- 2 **Appointment letters made mandatory** for every employee - a significant formalisation measure
- 3 **Annual free health examination** for workers above a prescribed age in specified establishments
- 4 **A single registration, single licence and single return** replacing multiple filings
- 5 **Inspector-cum-Facilitator** with **web-based randomised** inspection
- 6 **Contract labour** - a **single all-India licence** for contractors, and the Code prohibits contract labour in **core activities** subject to exceptions
- 7 **Inter-state migrant workers** - a **portability** of benefits, an **annual journey allowance** to the home State, and a **toll-free helpline**
- 8 **Women permitted in all establishments and all types of work**, including at night, **with consent** and prescribed safeguards

IMPORTANT · AN UNRESOLVED ITEM - STATED HONESTLY RATHER THAN GUESSED

The 2023 paper asked: "**Under the provisions of the Mines Act, 1952, the Special Officer can enter a mine for the purpose of... after giving notice of...**"

I have not been able to verify this notice period against a primary source, and I am not going to print a figure I cannot stand behind. A wrong number learned confidently is worse than a known gap, because you will carry it into the exam and into your work.

What to do: read the relevant section of the bare Mines Act, 1952 before the RT and write the figure into this box yourself. If you cannot find it, accept the loss of one mark - it is a single mark, and the cost of guessing is that you may also mislearn a neighbouring provision.

This is the only figure in the whole of Module 2 left deliberately blank. Everything else has been verified.

CHECK YOURSELF · SESSION 5

- 1 What does the Third Schedule contain? Who must notify, and who else must report?
- 2 Name six notifiable diseases, including the three classic Indian ones and their causes
- 3 Distinguish the contents of the First, Second and Third Schedules
- 4 When must notice of an accident be given?
- 5 How many Acts does the OSH Code subsume? Name six
- 6 Name five changes the OSH Code made
- 7 Which single figure in this module is deliberately left blank, and why?

6. Graded Practice - 45 Questions ★★★★★

6.1 Level 1 - Recognition

Q. 1. Under the Factories Act, 1948, premises are a factory where the number of workers working with the aid of power is

(a) 10 or more (b) 20 or more (c) 30 or more (d) 40 or more

Q. 2. Under the Factories Act, 1948, without the aid of power the threshold is

(a) 10 or more (b) 20 or more (c) 30 or more (d) 50 or more

Q. 3. Under the OSH Code, 2020, the factory thresholds are

(a) 10 with power and 20 without (b) 20 with power and 40 without (c) 30 with power and 60 without (d) 40 with power and 80 without

Q. 4. The overcrowding standard per worker is

(a) 9.9 cubic metres (b) 11.2 cubic metres (c) 14.2 cubic metres (d) 16.5 cubic metres

Q. 5. Drinking water points must be located away from any washing place or latrine by at least

(a) 3 metres (b) 6 metres (c) 9 metres (d) 12 metres

Q. 6. A canteen is required where the number of workers ordinarily employed exceeds

(a) 150 (b) 250 (c) 500 (d) 1,000

Q. 7. Shelters, rest rooms and lunch rooms are required where the number of workers exceeds

(a) 100 (b) 150 (c) 250 (d) 500

Q. 8. An ambulance room is required where the number of workers is

(a) 250 or more (b) 500 or more (c) 750 or more (d) 1,000 or more

Q. 9. A Welfare Officer is required where the number of workers is

(a) 250 or more (b) 500 or more (c) 1,000 or more (d) 1,500 or more

Q. 10. A Safety Officer is required where the number of workers is

(a) 250 or more (b) 500 or more (c) 1,000 or more (d) 2,000 or more

Q. 11. Under the Factories Act, a creche is required where the number of women workers ordinarily employed exceeds

(a) 20 (b) 30 (c) 50 (d) 100

Q. 12. One first-aid box must be provided for every

(a) 100 workers (b) 150 workers (c) 250 workers (d) 500 workers

Q. 13. The maximum weekly hours for an adult worker are

(a) 40 (b) 44 (c) 48 (d) 54

Q. 14. The maximum daily hours for an adult worker are

(a) 8 (b) 9 (c) 10 (d) 12

Q. 15. No period of work may exceed how many hours without a rest interval?

(a) four (b) five (c) six (d) seven

Q. 16. The spread-over limit in a day is

(a) 9 hours (b) 10 hours (c) 10.5 hours (d) 12 hours

Q. 17. No worker may work more than how many days consecutively without a holiday?

(a) six (b) seven (c) ten (d) twelve

Q. 18. Overtime under the Factories Act is payable at

(a) one and a half times the ordinary rate (b) twice the ordinary rate (c) two and a half times (d) three times

Q. 19. Annual leave with wages accrues to an adult at the rate of one day for every

(a) 15 days worked (b) 18 days worked (c) 20 days worked (d) 25 days worked

Q. 20. The Third Schedule to the Factories Act, 1948 contains

(a) a list of industries involving hazardous processes (b) permissible limits of exposure to toxic substances (c) a list of notifiable occupational diseases (d) the forms of registers to be maintained

Q. 21. The OSH Code, 2020 subsumes how many Acts?

(a) Four (b) Nine (c) Thirteen (d) Twenty-nine

6.2 Level 2 - Understanding

Q. 22. Which one of the following is expressly excluded from the definition of a factory?

(a) A cold storage unit (b) A printing press (c) A mine subject to the Mines Act (d) A power generating station

Q. 23. Which one of the following is not a manufacturing process under the Act?

(a) Packing of goods (b) Preserving articles in cold storage (c) Pumping of sewage (d) Retail sale of finished goods over a counter

Q. 24. The occupier of a factory owned by a company is

(a) the company itself (b) any one of the directors (c) the factory manager only (d) the majority shareholder

Q. 25. Notice must be given to the Inspector before occupying premises as a factory at least

(a) 7 days in advance (b) 15 days in advance (c) 30 days in advance (d) 60 days in advance

Q. 26. In computing the overcrowding standard, space above what height is disregarded?

(a) 3 metres (b) 4.2 metres (c) 5 metres (d) 6 metres

Q. 27. Cooled drinking water in hot weather and sanitary-type latrines become compulsory above

(a) 150 workers (b) 250 workers (c) 500 workers (d) 1,000 workers

Q. 28. The hazardous-process provisions were inserted following

(a) the Bombay textile strike (b) the Bhopal gas disaster of 1984 (c) the Chernobyl accident (d) the recommendations of the Whitley Commission

Q. 29. A Safety Committee is required in a factory carrying on a hazardous process where the number of workers is

(a) 20 or more (b) 50 or more (c) 100 or more (d) 250 or more

Q. 30. The First Schedule to the Factories Act contains

(a) notifiable diseases (b) industries involving hazardous processes (c) permissible exposure limits (d) welfare facilities

Q. 31. A child below what age may not be employed in a factory?

(a) 12 years (b) 14 years (c) 15 years (d) 18 years

Q. 32. An adolescent may be employed in a factory only if he holds

(a) a school leaving certificate (b) a certificate of fitness from a certifying surgeon (c) the written consent of a parent (d) a licence from the Inspector

Q. 33. Under the Factories Act, 1948, women could not be required to work except between

(a) 6 a.m. and 6 p.m. (b) 6 a.m. and 7 p.m. (c) 7 a.m. and 7 p.m. (d) 8 a.m. and 8 p.m.

Q. 34. Under the Labour Codes, night work by women is

(a) prohibited absolutely (b) permitted with their written consent, subject to conditions on transport, safety, adequate numbers and POSH compliance (c) permitted only in continuous-process industries (d) permitted only with the Inspector's approval

Q. 35. A worker qualifies for annual leave with wages under the Factories Act if he has worked in the calendar year for

(a) 180 days (b) 200 days (c) 240 days (d) 300 days

Q. 36. Unavailed leave may be carried forward by an adult worker up to

(a) 20 days (b) 30 days (c) 40 days (d) 45 days

Q. 37. Silicosis is caused by exposure to

(a) asbestos fibre (b) cotton dust (c) free silica dust (d) carbon bisulphide

Q. 38. Byssinosis is associated with

(a) mining (b) cotton textile work (c) shipbreaking (d) electroplating

Q. 39. A medical practitioner attending a person he believes to be suffering from a notifiable disease must

(a) treat the patient and maintain confidentiality (b) report to the Chief Inspector, failure attracting a penalty (c) report to the occupier only (d) report only if the patient consents

6.3 Level 3 - Recruitment Test standard

Q. 40. Which one of the following is provided under the Third Schedule of the Factories Act, 1948?

(a) The list of industries involving hazardous processes (b) The permissible limits of exposure to chemical and toxic substances (c) The list of notifiable occupational diseases (d) The particulars of the register of adult workers

Q. 41. Match List-I with List-II and select the correct answer using the code below:

List-I (Facility or officer)	List-II (Threshold)
A. Canteen	1. 1,000 workers
B. Ambulance room	2. 150 workers
C. Safety Officer	3. 250 workers
D. Rest rooms and lunch rooms	4. 500 workers

(a) A-3 B-4 C-1 D-2 (b) A-4 B-3 C-1 D-2 (c) A-3 B-1 C-4 D-2 (d) A-2 B-4 C-1 D-3

Q. 42. Consider the following statements:

- 1 A creche is required under the Factories Act where more than thirty women workers are ordinarily employed.
- 2 Under the Codes and the 2026 Rules, the creche requirement attaches at fifty employees, irrespective of how many are women.
- 3 The Maternity Benefit Act, as amended in 2017, also requires a creche at fifty or more employees.

Which of the above statements are correct?

(a) 1 and 2 only (b) 2 and 3 only (c) 1 and 3 only (d) 1, 2 and 3

Q. 43. The spread-over of ten and a half hours may be extended to twelve hours

(a) at the employer's discretion (b) with the written permission of the Chief Inspector (c) only in hazardous-process factories (d) only with the worker's consent

Q. 44. Which one of the following pairs is not correctly matched?

(a) Overcrowding - 14.2 cubic metres per worker (b) Drinking water distance from latrine - 6 metres (c) Self-acting machine traversing part clearance - 45 centimetres (d) Welfare Officer - 1,000 workers

Q. 45. Consider the following statements about the OSH Code, 2020:

- 1 It subsumes thirteen Acts, including the Factories Act 1948 and the Mines Act 1952.
- 2 It made appointment letters mandatory for every employee.
- 3 It reduced the qualifying period for annual leave with wages from 240 days to 180 days.

Which of the above statements are correct?

(a) 1 and 2 only (b) 2 and 3 only (c) 1 and 3 only (d) 1, 2 and 3

7. Answers and Explanations

7.1 Level 1

Q	Ans	Explanation
1	a	10 or more with power under the 1948 Act
2	b	20 or more without power
3	b	20 and 40 - the Code doubled both figures
4	c	14.2 cubic metres per worker
5	b	6 metres
6	b	250 workers - canteen
7	b	150 workers - shelters, rest rooms and lunch rooms
8	b	500 or more - ambulance room
9	b	500 or more - Welfare Officer
10	c	1,000 or more - Safety Officer. Welfare at 500, Safety at 1,000
11	b	30 women workers under the 1948 Act
12	b	One box per 150 workers - a ratio , not a threshold
13	c	48 hours
14	b	9 hours
15	b	Five hours , then at least a half-hour interval
16	c	10.5 hours , extendable to 12 with the Chief Inspector's written permission
17	c	Ten days
18	b	Twice the ordinary rate
19	c	One day for every 20 days for an adult; 15 for a child
20	c	Notifiable occupational diseases - asked in 2023
21	c	Thirteen - the largest of the four Codes by Acts replaced

7.2 Level 2

Q	Ans	Explanation
22	c	A mine subject to the Mines Act is expressly excluded, along with a mobile armed-forces unit, a railway running shed, and a hotel, restaurant or eating place. Cold storage, printing and power generation are all manufacturing processes
23	d	Retail sale over a counter is not a manufacturing process. Packing, cold storage and pumping all are
24	b	Any one of the directors - the Act names a human being so that a duty can be personally enforced
25	b	15 days; 30 days for a seasonal factory
26	b	Space above 4.2 metres is disregarded
27	b	250 workers - the same figure as the canteen

Q	Ans	Explanation
28	b	The Bhopal gas disaster of 1984 ; the provisions were inserted by the 1987 amendment
29	b	50 or more workers in a hazardous-process factory
30	b	Industries involving hazardous processes. First hazardous industries, Second exposure limits, Third diseases
31	b	14 years - and note this matches Article 24
32	b	A certificate of fitness from a certifying surgeon, with a token referring to it
33	b	6 a.m. to 7 p.m.
34	b	Permitted with written consent , plus transport, safety, adequate numbers and POSH compliance. The principle is to remove the risk, not the women
35	c	240 days - the same figure as continuous service for retrenchment in Chapter F2.3
36	b	30 days for an adult; 40 for a child, the child being treated more generously
37	c	Free silica dust - mining, stone-crushing, foundry work
38	b	Cotton dust in textile mills
39	b	He must report to the Chief Inspector , and failure attracts a penalty . The duty overrides ordinary confidentiality because the purpose is to detect a workplace hazard affecting others

7.3 Level 3

Q	Ans	Explanation
40	c	Notifiable occupational diseases. This is the 2023 question. (a) is the First Schedule and (b) is the Second - the two distractors are the other two Schedules, which is exactly how the question was built
41	a	A-3 canteen 250; B-4 ambulance room 500; C-1 Safety Officer 1,000; D-2 rest rooms 150
42	d	All three correct. The shift from "thirty women " to "fifty employees " treats childcare as a parental rather than a women's need, and the fifty-employee creche appears in both the Codes and the Maternity Benefit Act as amended in 2017
43	b	Written permission of the Chief Inspector
44	d	A Welfare Officer is required at 500 , not 1,000. 1,000 is the Safety Officer. The other three pairs are correct
45	d	All three correct - thirteen Acts, mandatory appointment letters, and the leave qualifying period reduced from 240 to 180 days while the one-in-twenty accrual rate was retained

8. One-Minute Revision

ONE-MINUTE REVISION · CHAPTER F2.6 - FACTORIES, HEALTH, SAFETY AND WELFARE

FACTORY 10 with power / 20 without (1948) · **20 / 40** (OSH Code - **doubled**) · counts workers on **any day of the preceding 12 months** · **EXCLUDES** mine under the Mines Act, mobile armed-forces unit, railway running shed, hotel/restaurant/eating place

MANUFACTURING PROCESS is WIDE - making, altering, repairing, **packing**, cleaning, breaking up · **pumping** · **generating or transmitting power** · printing and bookbinding · **shipbuilding and breaking** · **COLD STORAGE**

OCCUPIER person with **ultimate control** · firm → any **partner** · company → any **DIRECTOR** · Government factory → the person appointed to manage · *the Act names a human being so a duty can be personally enforced*

APPROVAL site permission **deemed granted if not decided in 3 MONTHS** · notice to Inspector **15 DAYS** before occupying, **30 DAYS** for a seasonal factory · **Site Appraisal Committee** for hazardous-process location

HEALTH NUMBERS overcrowding **14.2 cubic metres** per worker, ignoring space above **4.2 m** · drinking water **6 m** from washing places and latrines · above **250 workers** - **cooled** drinking water and **sanitary-type** latrines

SAFETY dangerous parts **securely fenced** · machinery in motion - **trained adult male, tight-fitting clothing, name in register** · self-acting machine clearance **45 cm** · **excessive weights** prohibited · eye protection · **SAFETY OFFICER at 1,000**

HAZARDOUS PROCESSES inserted by the **1987 amendment** after **BHOPAL 1984** · Site Appraisal Committee · **compulsory disclosure - the right to know** · medical examination · **permissible exposure limits in the SECOND SCHEDULE** · **SAFETY COMMITTEE at 50 workers** · workers' right to warn of imminent danger

THE THREE SCHEDULES FIRST = hazardous **INDUSTRIES** · **SECOND** = permissible **EXPOSURE LIMITS** · **THIRD** = **NOTIFIABLE DISEASES** (asked 2023)

WELFARE TRIGGERS - CHANT THIS 150 rest rooms, shelters, lunch rooms · **250 CANTEEN** · **500 AMBULANCE ROOM** and **WELFARE OFFICER** · **1,000 SAFETY OFFICER** · **30 women CRECHE** (1948) · **50 employees creche** (Codes and Maternity Benefit 2017) · **1 first-aid box per 150** (a ratio) · washing, drying, sitting - **every factory**

HOURS AS A SENTENCE **48 a week** · **9 a day** · **5 hours then a half-hour break** · **10.5 spread-over (12 with the CHIEF INSPECTOR's written permission)** · **1 weekly holiday, substitution within 3 days either side** · **never more than 10 DAYS consecutively** · **overtime at TWICE** · **144 hours per quarter** · **no double employment on the same day**

YOUNG PERSONS AND WOMEN child under **14 prohibited** (matches **Art 24**) · **adolescent 14-18 needs a CERTIFICATE OF FITNESS and a token** · a fit young person of **15+** may be treated as an adult for hours · children **4.5 hours, no night work** · women **6 a.m. to 7 p.m.** under 1948 ·

Codes permit night work with WRITTEN CONSENT plus transport, safety, adequate numbers, POSH - *remove the risk, not the women*

LEAVE qualify with **240 DAYS** (OSH Code 180) · accrual **1 day per 20 days worked** for an adult, **1 per 15** for a child · carry forward **30 days** adult, **40 days** child · *the child is treated more generously on both counts* · **240 is the same figure as continuous service in F2.3**

NOTIFIABLE DISEASES manager notifies the **Chief Inspector**; a **medical practitioner** must also report, on **penalty** · lead, mercury, manganese, arsenic, phosphorus poisoning · benzene · chrome ulceration · **anthrax** · **SILICOSIS** - **free silica dust** · **ASBESTOSIS** - **asbestos fibre** · **BYSSINOSIS** - **cotton dust** · radiation manifestations · occupational **cataract** from infra-red · **compressed air illness** · **noise-induced hearing loss** · occupational asthma · contact dermatitis · pesticide poisoning

ACCIDENT NOTICE death, or injury preventing work for **48 HOURS or more** · **dangerous occurrences notified whether or not injury results**

OSH CODE - 13 ACTS Factories 1948 · Mines 1952 · Contract Labour 1970 · Inter-State Migrant 1979 · Building and Construction 1996 · Plantations 1951 · Dock Workers 1986 · plus working journalists, motor transport, beedi and cigar, sales promotion, cine workers, employment exchanges · **CHANGES:** thresholds **doubled** · **APPOINTMENT LETTERS MANDATORY** · annual **free health examination** · **single registration, licence and return** · **Inspector-cum-Facilitator** with web-based randomised inspection · **single all-India contractor licence** and contract labour barred in **core activities** · migrant workers get **portability, an annual journey allowance and a helpline** · women in **all establishments and all work**

LEFT DELIBERATELY BLANK the **Mines Act Special Officer notice period** asked in 2023 - unverified, so **not guessed**. Read the bare section and fill it in yourself

9. Five-Minute Wall Sheet

FIVE-MINUTE WALL SHEET · CHAPTER F2.6 - THE TRIGGER NUMBERS

FACTORY 10 / 20 old · 20 / 40 Code

TRIGGERS 150 rest rooms · **250** canteen · **500** ambulance room + **Welfare Officer · 1,000 Safety Officer · 30 women** creche (1948) · **50 employees** creche (Codes) · **50** Safety Committee (hazardous) · **1 first-aid box / 150**

WELFARE 500 · SAFETY 1,000 (welfare is cheaper, so it comes first)

HEALTH 14.2 m³ · above 4.2 m ignored · water 6 m from latrine · 250 for cooled water

HOURS 48 / 9 / 5-then-half-hour / 10.5 (12 by Chief Inspector) / 10 days max / OT twice / 144 a quarter

AGES 14 prohibited · 14-18 certificate of fitness · child 4.5 hours, no nights · women 6 a.m.-7 p.m. old, night with CONSENT under the Codes

LEAVE 240 days (Code 180) · 1 per 20 adult, 1 per 15 child · carry 30 / 40

SCHEDULES 1st HAZARDOUS INDUSTRIES · 2nd EXPOSURE LIMITS · 3rd NOTIFIABLE DISEASES

THREE CLASSIC DISEASES silicosis = silica · asbestosis = asbestos · byssinosis = cotton dust

ACCIDENT NOTICE at 48 HOURS' incapacity

HAZARDOUS PROVISIONS = 1987, AFTER BHOPAL 1984

OSH CODE = 13 ACTS · APPOINTMENT LETTERS MANDATORY · SINGLE LICENCE · THRESHOLDS DOUBLED

ACTION · NEXT IN THE FOUNDATION TRACK

Chapter F2.7 - Contract Labour, Inter-State Migrant Workers, Construction and Mines: the statutes that govern **indirect** employment, where the principal employer's secondary liability for provident fund contributions arises - which makes it one of the most directly relevant chapters in Module 2 to your daily work as an APFC.